DevDiscuss - Backend Project Documentation

1. Application Overview

DevDiscuss is a developer-focused discussion platform inspired by StackOverflow and GitHub Issues. Users can post programming problems, submit solutions, upvote content, follow other users, and receive notifications. The system is built using Next.js (App Router), PostgreSQL, and Drizzle ORM.

2. Core Functionalities

- User Authentication (Register, Login, JWT-based sessions)
- Create, Update, Delete Problems
- Post Solutions to Problems
- Upvote Problems and Solutions (Unique vote per user)
- Mark Solution as Accepted (Problem owner only)
- Follow / Unfollow Users
- Notifications system
- Search, Filter, and Pagination support (future expansion)

3. Full Backend File Structure

```
devdiscuss/
■
■■■ app/
■   ■■■ api/
■   ■   ■■■ auth/
■   ■   ■   ■■■ register/route.ts
■   ■   ■   ■■■ login/route.ts
■   ■   ■   ■■■ me/route.ts
■   ■   ■
■   ■   ■■■ problems/
■   ■   ■   ■■■ route.ts
■   ■   ■   ■■■ [id]/route.ts
■   ■   ■   ■■■ [id]/vote/route.ts
■   ■   ■   ■■■ [id]/solutions/route.ts
■   ■   ■
■   ■   ■■■ solutions/
■   ■   ■   ■■■ [id]/route.ts
■   ■   ■   ■■■ [id]/vote/route.ts
■   ■   ■   ■■■ [id]/accept/route.ts
■   ■   ■
■   ■   ■■■ comments/
■   ■   ■   ■■■ [id]/route.ts
■   ■   ■
■   ■   ■■■ users/
■   ■   ■   ■■■ route.ts
■   ■   ■   ■■■ [id]/route.ts
■   ■   ■   ■■■ [id]/follow/route.ts
■   ■   ■   ■■■ [id]/followers/route.ts
■   ■   ■   ■■■ [id]/following/route.ts
■   ■   ■
■   ■   ■■■ notifications/
■   ■   ■   ■■■ route.ts
■   ■   ■   ■■■ [id]/read/route.ts
```

```
■   ■   ■   ■■■ read-all/route.ts
■   ■   ■
■   ■   ■■■ leaderboard/route.ts
■   ■   ■■■ trending/route.ts
■   ■
■   ■■■ layout.tsx
■
■■■ db/
■   ■■■ index.ts
■   ■■■ schema/
■   ■   ■■■ users.ts
■   ■   ■■■ problems.ts
■   ■   ■■■ solutions.ts
■   ■   ■■■ votes.ts
■   ■   ■■■ follows.ts
■   ■   ■■■ notifications.ts
■   ■
■   ■■■ migrations/
■
■■■ lib/
■   ■■■ jwt.ts
■   ■■■ auth.ts
■   ■■■ validators.ts
■   ■■■ utils.ts
■
■■■ middleware.ts
■■■ drizzle.config.ts
■■■ .env
■■■ package.json
```

4. Architecture Notes

- App Router based API architecture. - Drizzle ORM for type-safe SQL queries. - PostgreSQL as primary database. - JWT stored in HTTP-only cookies for authentication. - Middleware protects private API routes. - Clean separation: API layer, DB layer, and business logic layer.